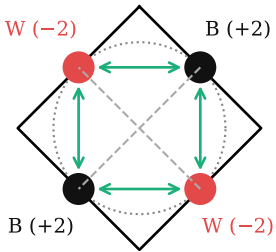
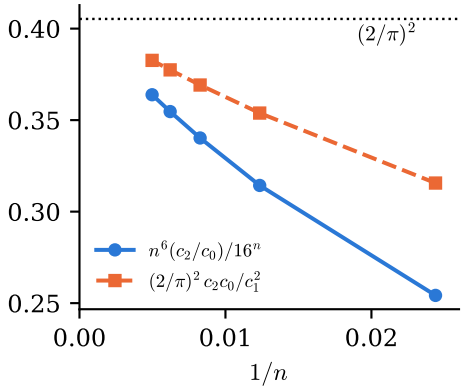
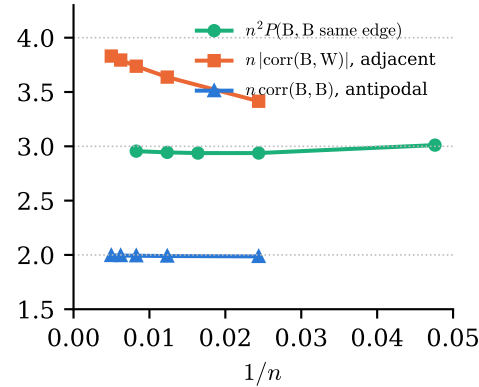
(a) dominant $k=2$ configuration



(b) weight of two pairs



(c) residual interactions